

Qian Jun BEH

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EDUCATION

National University of Singapore

Aug 2021 – Aug 2025

Bachelor of Science with Honours (Distinction) in Physics, Second Major in Data Analytics, Special Programme in Science (SPS).

WORK EXPERIENCE

Research Engineer, Singapore Institute of Manufacturing Technology (A*STAR SIMTech)

Aug 2025 – Present

- Developed supervised and semi-supervised fine-tuning techniques on the time-series foundation model for CNC fault detection and tool-wear prediction tasks.
- Developing context-aware time-series generation methods for anomalous industrial data to support rare-event modelling and predictive-maintenance experimentation.
- Achieved up to ~ 0.90 F1 for fault classification with supervised fine-tuning and around 15% MAPE for tool-wear regression with semi-supervised fine-tuning.
- Built end-to-end fine-tuning and evaluation pipelines on an HPC cluster (Slurm, Apptainer) to support large-scale experimentation.
- Co-authored an accepted abstract at CLF 2026 (Conference on Learning Factories) presenting a fine-tuning pipeline for factory learners.
- Collaborated with multiple industrial partners to translate research and AI solutions into real-world predictive-maintenance deployments.

Research Intern, Institute of High Performance Computing (A*STAR IHPC)

Mar 2025 – Jul 2025

- Project: Meta-PINN for the Grad-Shafranov Equation: Forward Prediction and Inverse Parameter Inference*
- Implemented JAX-based Meta-PINNs to solve the Grad-Shafranov equation for fusion plasma equilibrium.
- For forward prediction, matched analytical solutions with mean squared error on the order of 10^{-11} .
- For inverse parameter inference, recovered equilibrium parameters with mean squared error on the order of 3×10^{-5} .
- Currently working on the write-up for journal submission.

Machine Learning Intern, Institute of Mental Health

Jul 2024 – Feb 2025

- Developed a generative modelling pipeline using a Variational Autoencoder (VAE), achieving 20% improvement in prediction accuracy compared to baseline models; findings showcased at the Singapore Health and Biomedical Congress 2024 (SHBC 2024).
- Leveraged Ray, a parallel processing framework, on a high-performance computing (HPC) cluster for efficient hyperparameter tuning, enhancing model performance and scalability.
- Created the Nutrients Pipeline (nPipeline), utilising language embedding models and large language models (LLMs) to match patient data with studies from the genetic database, automating the extraction of relevant summary statistics.
- Developed computational methods to determine the optimal sequence of tests, improving the efficiency and accuracy of real-life medical diagnostics, e.g. the diagnosis of mental health disorders.

Other Experience: Student Helper for Introduction to Data Science; Student Research Assistant analysing HRV and EDA time-series data.

RELEVANT PROJECTS

CS3244 Machine Learning Project: UCI Human Activity Recognition (HAR)

Oct 2024 – Dec 2024

- Experimented with various machine learning models, including Random Forest, XGBoost, kNN, and LSTM, with the best model achieving a test F1 score of > 0.92 .
- Implemented parallelised hyperparameter tuning using Ray and applied LIME for interpretable model explanations.
- Utilised Git for version control and collaboration; project repository: github.com/rachelloo23/CS3244.

SNP-s Analysis Variational Autoencoder

Jan 2023 – Apr 2023

- Developed a machine learning pipeline to predict toxicity induced by chemotherapy drugs in patients, using generative models like Variational Autoencoder (VAE) alongside Random Forest, Gradient Boosting, Artificial Neural Network (ANN), and Logistic Regression.
- Preprocessed a vast dataset containing cancer patients' information, involving over 700,000 columns of SNPs.

PUBLICATIONS & CONFERENCES

- Journal:** Qian Jun Beh et al., "Autoencoding-Assisted Quantum Cloning Machine," *Entropy*, 28(5), 563, 2026. DOI: 10.3390/e28050563.
- Conference for Learning Factories 2026:** "From Signals To Decisions: A Pedagogical Pipeline for Tool-Wear Prediction Using Few-Shot Learning."
- Singapore Health and Biomedical Congress 2024:** "A Genetic Approach to Predicting Nutrient Deficiencies Using Variational Autoencoder (VAE)."

ADDITIONAL INFORMATION & SKILLS

Programming languages: Java, Python, Matlab, R, HTML, CSS, JavaScript, SQL

Skills: Machine Learning, Data Analysis, Statistical Analysis, Data Visualisation, Optimisation, Git, LaTeX